

# LinkedIn Job Trend Analysis

## Skill Demand & City-wise Job Trend Report

### 1. Objective

This project analyzes job posting data to understand which skills are most in demand and which cities have the highest concentration of job openings. The goal is to identify skill-demand trends across roles and locations to guide career and hiring decisions.

### 2. Tools Used

- Python (Pandas, Matplotlib)
- Data generation & cleaning
- Visualization (bar charts)

### 3. Methodology

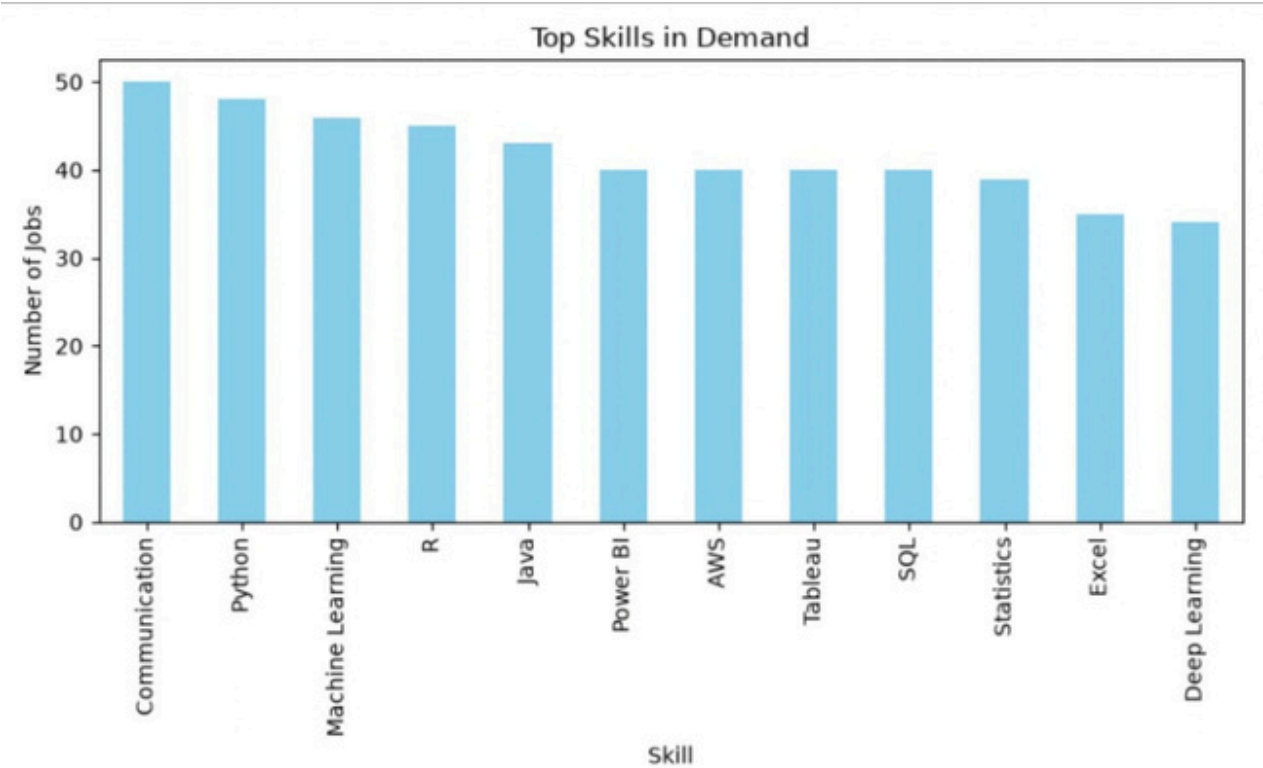
- Generated a structured job-postings dataset with Job Title, City, and Skill fields.
- Cleaned and loaded the dataset using Pandas.
- Aggregated counts of skills and cities to identify top trends.
- Built a Skill vs Role matrix using a cross-tabulation.
- Visualized results using bar charts for clear comparison.

### 4. Key Results

| Metric                      | Result                                                                         |
|-----------------------------|--------------------------------------------------------------------------------|
| Total Job Postings Analyzed | 500                                                                            |
| Most In-Demand Skill        | Communication                                                                  |
| Highest Job-Demand City     | Chennai                                                                        |
| Roles Covered               | Data Analyst, Data Scientist, Software Engineer, Business Analyst, ML Engineer |
| Cities Covered              | Bangalore, Mumbai, Delhi, Hyderabad, Pune, Chennai                             |

## 5. Top Skills in Demand

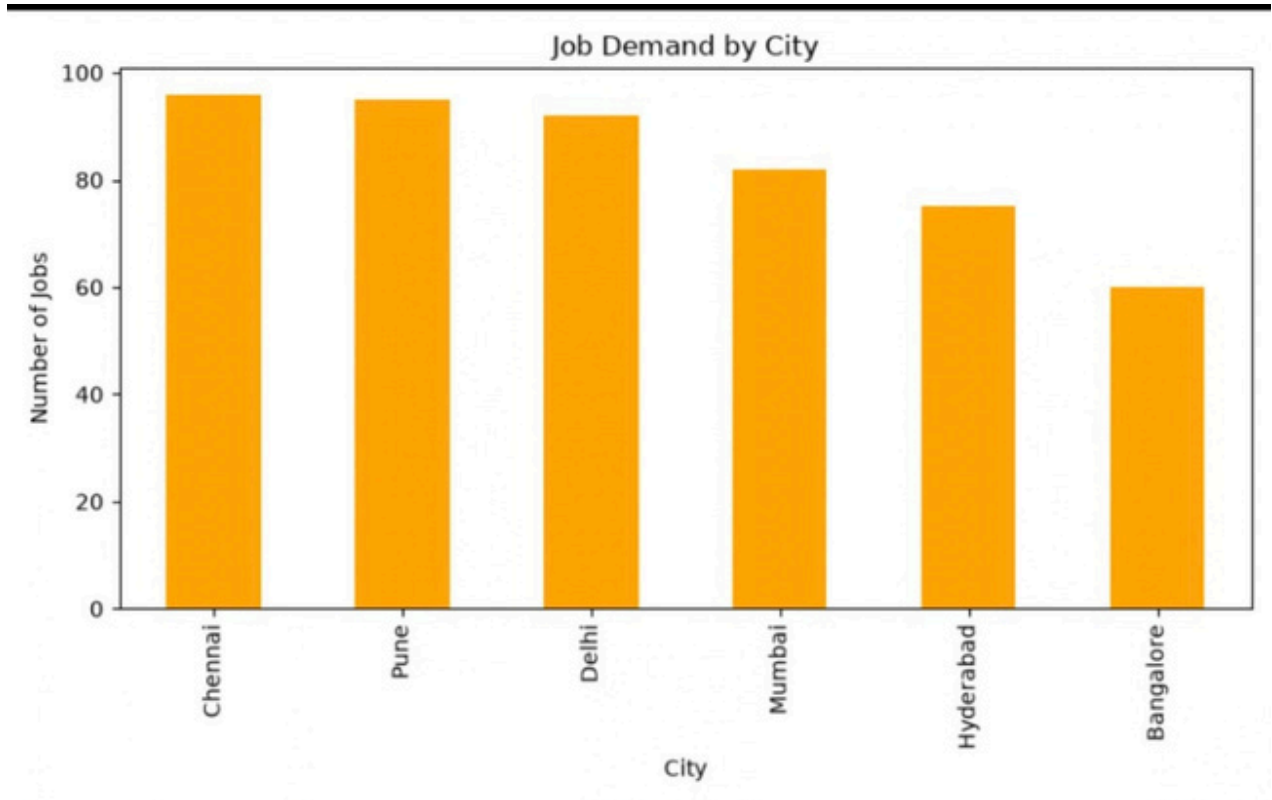
The chart below shows the frequency of each skill across all job postings analyzed.



Communication, Python, and Machine Learning emerge as the most frequently required skills, followed closely by R and Java. This suggests that alongside technical skills, soft skills like communication remain highly valued by employers.

## 6. Job Demand by City

The chart below shows how job openings are distributed across major cities.



Chennai, Pune, and Delhi show the highest job demand, while Bangalore, despite its reputation as a tech hub, shows comparatively fewer postings in this sample. This highlights that job opportunities are spreading well beyond traditional IT hubs.

## 7. Skill vs Role Matrix

A cross-tabulation of Job Title against Skill was generated (saved as `skill_vs_role_matrix.csv`) to understand which skills are most associated with which roles. This helps in mapping specific skill requirements to specific job titles for targeted upskilling.

## 8. Recommendation

Based on the analysis, candidates should prioritize strengthening their Communication and Python skills, as these show the highest demand across job postings. Additionally, job seekers may consider exploring opportunities in Chennai, which currently shows the highest concentration of job openings among the cities analyzed.

## 9. Deliverables

- `linkedin_jobs.csv` — Job postings dataset
- `top_skills.png` — Skill demand visualization
- `jobs_by_city.png` — City-wise demand visualization
- `skill_vs_role_matrix.csv` — Skill vs Role cross-tabulation
- This report (`LinkedIn_Job_Trend_Report.docx`)